$\therefore$ Area of the entire mat

= area of the I rectangle + area of the II rectangle

$= l_1 \times b_1 + l_2 \times b_2$

$= 5 \times 2 + 9 \times 2 = 10 + 18 = 28$ *sq.feet*

Cost per sq. foot = ₹ 20

$\therefore$ The total cost of the entire mat = 28 × ₹ 20 = ₹ 560.



Fig. 2.28

Try these

In the above example split the given mat into two trapeziums and verify your answer.

Example 2.10

Find the area of the shaded region in the square of side 10 *cm* as given in the Fig. 2.29. $\left(\pi = \frac{22}{7}\right)$

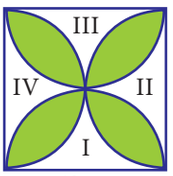


Fig. 2.29

Solution:

Mark the unshaded parts of the given figure as I, II, III and IV

Area of the I and III parts = Area of the square – Area of 2 semicircles

$$= a^2 - \left(2 \times \frac{1}{2}\pi r^2\right)$$

$$= 10 \times 10 - \frac{22}{7} \times 5 \times 5 \quad = 100 - 78.57 = 21.43 \ cm^2.$$

Similarly, the area of the II and IV parts = 21.43 cm^2 .

$\therefore$ Area of the unshaded parts (I, II, III and IV)

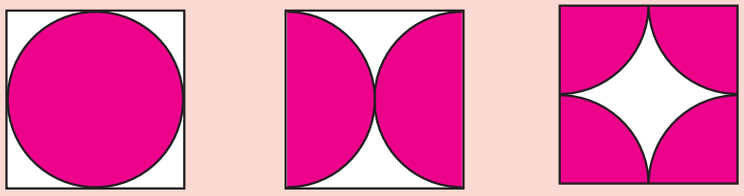
= 21.43 ×2 = 42.86 cm^2 (approximately)

$\therefore$ Area of the shaded part = area of the square – area of the unshaded parts

= 100 – 42.86 = 57.14 cm^2 (approximately)

DO YOU KNOW?

1. The area of the unshaded regions in each of the squares of side *a* units are the same in all the cases given below.



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